

Each (QKP) instance correspond to a file containing the following informations:

- the reference of the instance (r_10_100_13 in the following example)
- the number of variables (n) (10 in the following example)
- the linear coefficients (c_i) of the objective function (91 78 22 4 48 85 46 81 3 26)
- the quadratic coefficients (c_{ij}) of the objective function
- a blank line
- 0 if the constraint is of type $\leq$ (i.e. always since we are considering (QKP) instances) and 1 if the constraint is an equality constraint
- the capacity of the knapsack (145)
- the coefficients of the capacity constraints (weights, a_i) (34 33 12 3 43 26 10 2 48 39)
- some comments

Here is an example concerning a 10-variable instance:

```
10
91 78 22 4 48 85 46 81 3 26
55 23 35 44 5 91 95 26 40
92 11 20 43 71 83 27 65
7 57 33 38 57 63 82
100 87 91 83 44 48
69 57 79 89 21
9 40 22 26
50 6 7
71 52
17

0
145
34 33 12 3 43 26 10 2 48 39
```

This example corresponds to the following (QKP) instance:

$$(QKP) \left\{ \begin{array}{l} \max \quad 91x_1 + 78x_2 + 22x_3 + 4x_4 + 48x_5 + 85x_6 + 46x_7 + 81x_8 + 3x_9 + 26x_{10} \\ \quad + 55x_1x_2 + 23x_1x_3 + 35x_1x_4 + 44x_1x_5 + 5x_1x_6 + 91x_1x_7 + 95x_1x_8 + 26x_1x_9 + 40x_1x_{10} \\ \quad + 92x_2x_3 + 11x_2x_4 + 20x_2x_5 + 43x_2x_6 + 71x_2x_7 + 83x_2x_8 + 27x_2x_9 + 65x_2x_{10} \\ \quad + 7x_3x_4 + 57x_3x_5 + 33x_3x_6 + 38x_3x_7 + 57x_3x_8 + 63x_3x_9 + 82x_3x_{10} \\ \quad + 100x_4x_5 + 87x_4x_6 + 91x_4x_7 + 83x_4x_8 + 44x_4x_9 + 48x_4x_{10} \\ \quad + 69x_5x_6 + 57x_5x_7 + 79x_5x_8 + 89x_5x_9 + 21x_5x_{10} \\ \quad + 9x_6x_7 + 40x_6x_8 + 22x_6x_9 + 26x_6x_{10} \\ \quad + 50x_7x_8 + 6x_7x_9 + 7x_7x_{10} \\ \quad + 71x_8x_9 + 52x_8x_{10} \\ \quad + 17x_9x_{10} \\ \\ \text{s.t.} \quad \left| \begin{array}{l} 34x_1 + 33x_2 + 12x_3 + 3x_4 + 43x_5 + 26x_6 + 10x_7 + 2x_8 + 48x_9 + 39x_{10} \leq 145 \\ x_i \in \{0, 1\} \quad \forall i \in \{1, \dots, n\} \end{array} \right. \end{array} \right.$$